Toulouse School of Economics, 2013-2014
Macroeconomics 1 – Franck Portier

Problem set 5: a matching model

We consider the following matching model. The matching function is given by

$$m(u,v) = \frac{uv}{u+v}. \tag{1}$$

There is free entry in vacancy posting and the cost of posting a vacancy per unit of time is equal to c. Wages are set by intertemporal Nash bargaining so that

$$V_e = V_u + \varphi W, \tag{2}$$

and

$$J = (1-\varphi)W \tag{3}$$

The job destruction rate is s and the real interest rate is r.

1 – What is the function $q(\theta)$? Check that $m(u,v)$ has constant returns to scale and is concave in each of its arguments. What is the elasticity of m with respect to u as a function of θ?

2 – Show that the Beveridge curve has the following equation:

$$\theta = \frac{s(1-u)}{u - s(1-u)} \tag{4}$$

3 – Show that the equilibrium value of θ must satisfy:

$$y/c = \frac{r+s}{1-\varphi}(1+\theta) + \frac{\varphi}{1-\varphi}\theta \tag{5}$$

4 – What is the steady-state unemployment rate u? How does it depend on r, s, c, y, and φ? Why?

5 – Show that the following relationship exists between the wage w and θ :

$$w = \frac{\varphi c}{1-\varphi}[(r+s)(1+\theta)+\theta] \tag{6}$$

6 – How do wages depend on r, s, c, φ, θ according to this formula? Explain?

7 – What would be the observed correlation between wages and unemployment if the driving force for changes in unemployment was:

- changes in productivity ?
- changes in the discount rate?
- changes in the cost of posting vacancies?
- changes in workers' bargaining power?

8 – Write down the central planner's problem, the Hamiltonian, and the corresponding first-order conditions.

9 – Show that along the optimal path the marginal value of an extra job λ is related to θ according to:

$$\lambda = c(1+\theta)^2. \tag{7}$$

Explain the economic intuition for why it increases with c and θ.

10 – Show that in steady state the optimal value of θ must satisfy

$$y/c = (r+s)(1+\theta)^2 + c\theta^2 \tag{8}$$

11 – What relationship must hold among the model's parameters for (8) to have the same solution as (5) ?

12 – Check that this is equivalent to

$$\varphi = \frac{\theta}{1+\theta}, \tag{9}$$

with θ being the common solution to these two equations.

13 – How does this last condition relate to the Hosios efficiency condition?